Bacterial Extracellular Vesicles (EVs) Extraction

Quick Overview

Parameter	Information
Protocol ID	EV-Extraction-001
Version	1.0
Method Owner	Haining Huang
Estimated Hands-on Time	~4–6 h (excluding the unattended centrifugation and the 2 h ultracentrifugation run)
Total Duration	~6.5–9 h of bench time; usually spread over 1–3 days with optional overnight 4°C holds
Difficulty Level	☐ Basic ☐ Intermediate ☑ Advanced
Training Required	☑ Yes ☐ No
Biosafety Level	Depends on the strain used; in all cases, work must comply with the applicable institutional laboratory safety regulations
Typical Throughput	Max ~5 L of culture processed each time
Sample types	Bacterial culture supernatant (e.g., *E. coli* or *P. putida* liquid culture)
Output	Purified bacterial extracellular vesicles (EVs), resuspended in ~1 mL PBS
Related Protocols	EV-Purification-001 (Density Gradient Purification — see Optional Step)

Method Information

Method Title	SOP for Bacterial Extracellular Vesicles (EVs) Extraction
Author(s)	Haining Huang
Date Created	2026-08-28
Last Updated	2026-10-21
Reviewed By	Chika Ejikeugwu
Approved By	

Purpose

To isolate and enrich extracellular vesicles (EVs) from bacterial culture supernatant through sequential centrifugation, filtration, concentration, and ultracentrifugation, yielding an EV preparation suitable for downstream functional validation assays. If proteomic analysis is required, further purification by density gradient can be performed (Optional Step).

Typical Starting Material

Typically, *E. coli* MG1655 carrying plasmid pKJK5 or *P. putida* KT2440 carrying plasmid RSF1010 is cultured in LB medium at 37°C for 16 h at 150 rpm until it reaches an OD600 of approximately 1.5. Approximately 2 L of culture is processed per batch.

Scope

Suitable for

- EV isolation from bacterial liquid culture supernatant (broth culture)
- Sample volumes from <1 L (dead-end filtration can be used) up to ~5 L (using the TFF system)
- EV preparations intended for downstream functional validation assays

Not suitable for

- Non-liquid/solid samples

Safety Considerations

Hazards

Biological: Bacterial culture supernatant may contain live bacterial cells or residual pathogens prior to filtration; handle at the appropriate biosafety level

Chemical: 0.01 M NaOH (mildly alkaline, used for TFF system storage) — avoid contact with skin and eyes; wear safety goggles

Flammable: Ethanol (70–75%, lab-grade; used to clean ultracentrifuge tubes and filter inserts) — keep away from open flames and other ignition sources

Physical: Ultracentrifugation at 100,000 × g — if a tube is underfilled, the vacuum drawn during the run may crush the thin-wall tube (see Critical Control Points below); observe standard mechanical safety when operating the centrifuge/ultracentrifuge

Required PPE

- ☑ Lab coat
- ☑ Gloves
- ☑ Safety goggles (when handling NaOH or ethanol)

Waste Disposal

Biological waste: Centrifuge bottles and ultracentrifuge tubes, filter membranes and TFF cassettes that have been in contact with bacterial culture are decontaminated before reuse or disposal (see “Decontamination of Reusable Bottles, Tubes and Glassware” above). Single-use items — filter membranes, syringe filters and pipette tips — are discarded as biological waste; reusable items are collected centrally for decontamination and washing.

Chemical waste: 0.01 M NaOH solution — dispose of per institutional chemical waste guidelines

Decontamination of Reusable Bottles, Tubes and Glassware

Bottles, tubes and glassware that have been in contact with bacterial culture are decontaminated by **one** of the two routes below before washing. Use one route only: do not autoclave items that have already been chemically disinfected, and avoid autoclaving the same item repeatedly — the plastic degrades with every cycle.

Route A — Autoclaving. Separate the bottles from their caps, bag them and autoclave, then wash. This is the most thorough option, but polypropylene bottles and thin-wall ultracentrifuge tubes tolerate only a limited number of cycles; keep a record of the cycles for items that are reused often.

Route B — Chemical disinfection. Immerse the items in 2% (v/v) Wofasteril (peracetic acid) for 1–2 h, then rinse thoroughly with water and wash as usual. Use this route for items that would otherwise have to be autoclaved repeatedly.

Materials and Reagents

Equipment

Equipment	Model	Location
1 L centrifuge bottles	Thermo Scientific	E27/K21 in building 4.1 (UFZ)
SORVALL LYNX 6000 Centrifuge	Thermo Scientific Inc., Waltham, MA, USA	K21 in building 4.1 (UFZ)
Vacuum pump	Laboratory diaphragm vacuum pump (any model suitable for filtration)	E29 in building 4.1 (UFZ)
Vacuum filtration flask	e.g., Büchner/side-arm filter flask, used with the vacuum pump	E29 in building 4.1 (UFZ)
Concentrator tubes	Amicon® Ultra-15, 100 kDa MWCO	E29 in building 4.1 (UFZ)
Tangential Flow Filtration (TFF) system	Interchangeable cassettes: Sartorius 0.2 µm cassette (Step 2, filtration) and Millipore 100 kDa cassette (Step 3, Method B)	One in K29 in building 4.1 (UFZ), one in BTU
Analytical balance	precision 0.01 g	E29 in building 4.1 (UFZ)
Ultracentrifuge + rotor	Beckman Coulter Optima XP-80 ultracentrifuge with SW 32 Ti swinging-bucket rotor and buckets (#1–#6)	Lab 221 in building 4.1 (UFZ)
Ultracentrifuge tubes	Open-Top Thinwall PP, 38.5 mL, 25×89 mm, Cat. No. 326823	E29 in building 4.1 (UFZ)

Materials and Reagents

Reagent/Material	Specification	Supplier	Storage
1× PBS (stock)	Not filtered; general-purpose stock solution	Prepared in-house	Room temperature (RT)
1× PBS (filtered, cold)	Sterile, 0.2 µm filtered, pre-chilled — used for resuspension steps (e.g., Step 4.7)	Prepared in-house	4°C
0.01 M NaOH (working solution)	0.01 M (in DI water)	Prepared in-house	RT
PBS salts	NaCl, KCl, Na_2HPO_4 (anhydrous), KH_2PO_4 — analytical grade; used to prepare the 1× PBS stock (Table 2)	Any supplier	RT
NaOH (solid)	Pellets, analytical grade; used to prepare the 0.01 M NaOH solution	Any supplier	RT
Ethanol	70–75% (v/v), lab-grade; used to clean ultracentrifuge tubes and concentrator filter inserts	Any supplier	RT — flammable; store per institutional rules

Reagent/Material	Specification	Supplier	Storage
Deionized (DI) water	Large volume consumable; used for rinsing the TFF system, concentrator tubes, and ultracentrifuge tubes	Available in the lab	RT
Media bottle	5 L, glass	Any supplier	RT
Erlenmeyer flask	1 L, 2 L or 3 L	Any supplier	RT
Centrifuge tubes	50 mL	Any supplier	RT
Filter membranes (dead-end filtration)	PES, 0.8 / 0.45 / 0.2 µm, 47 mm diameter — see Ordering Information	Cytiva (Pall) Supor or GVS (or any supplier)	RT
Syringe filters	0.2 µm, sterile	https://www.sigmaaldrich.com/DE/de/product/mm/slgpr33rs https://www.fishersci.de/shop/products/sterile-pes-syringe-filter/15206869	RT
Syringes	20 mL / 50 mL	Any supplier	RT
EP tubes	1.5 mL/2 mL	Thermo Fisher	RT
General labware	Pipettes, motorized pipette filler, tube racks, micropipettes (various volumes), sterile syringes	Any supplier	RT

Reagent Preparation

Solution/Buffer 1 — 1× PBS (RT stock)

Table 1. Solution/Buffer 1 — 1× PBS (RT stock)

Composition	1× PBS, autoclaved (filtration optional)
Preparation	See Table 2
Storage	Room temperature (RT)
Stability	`Stable for ≥1 year at RT`

Table 2. Composition of 1× PBS (0.01 M, pH 7.2)

Component	Amount per 1 L
NaCl	8 g
KCl	0.2 g
Na_2HPO_4	1.44 g
KH_2PO_4	0.24 g
Solvent	Double-distilled water (ddH_2O), make up to 1 L
pH	Adjust to 7.2

Solution/Buffer 2 — 1× PBS (filtered, cold)

Table 3. Solution/Buffer 2 — 1× PBS (filtered, cold)

Composition	1× PBS, sterile, 0.2 µm filtered
Preparation	Filter the PBS stock through a 0.2 µm filter
Storage	4°C
Stability	Stable

Solution/Buffer 3 — 0.01 M NaOH

Table 4. Solution/Buffer 3 — 0.01 M NaOH

Composition	0.01 M NaOH in DI water
Preparation	Dissolve 0.4 g of NaOH in 1 L of DI water.
Storage	Room temperature (RT)
Stability	Used to flush/store the TFF system between runs to inhibit bacterial growth; very stable

Procedure

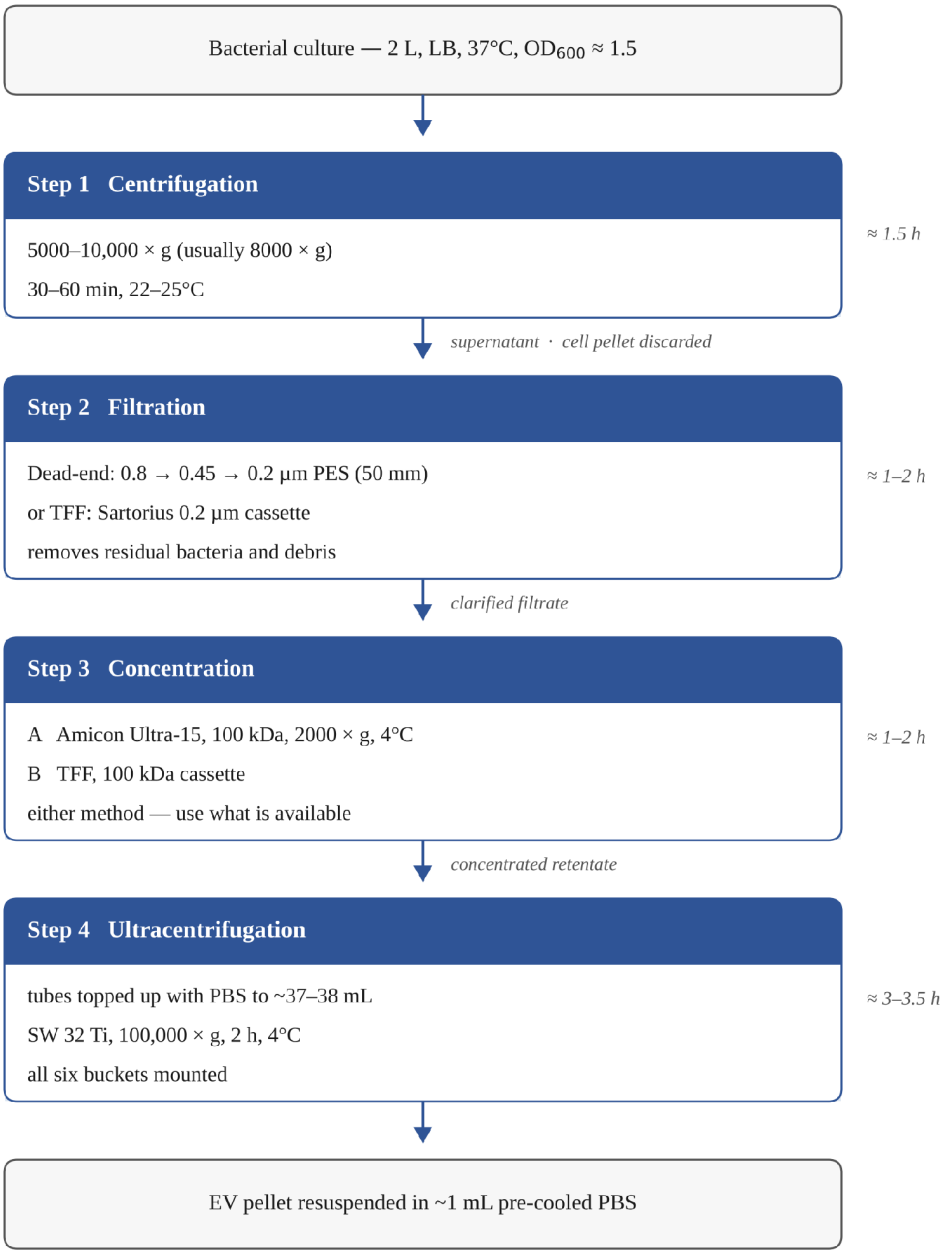


Figure 1. Workflow overview of the bacterial EV extraction procedure

Approximate time for each step is shown on the right. Steps 1–2 remove bacteria and debris; Steps 3–4 enrich the vesicles.

Before Starting — Checklist

- ☐ Required training completed
- ☐ Reagents prepared (1× PBS, 0.01 M NaOH)
- ☐ Equipment available (TFF system, ultracentrifuge and rotor booked/confirmed)
- ☐ Calibration verified (analytical balance)
- ☐ Sample IDs assigned

Step 1: Centrifugation (Bacterial Removal)

Note: Because this is a bacterial culture (unlike, e.g., soil or environmental samples), the starting density of the culture is relatively uniform, so balancing between bottles is straightforward. Bottles must still be balanced carefully before every run (see Critical Control Points).

1. Pour the bacterial culture into 1 L centrifuge bottles.
2. Strictly balance the bottles using a scale or by matching the liquid volume meniscus.
3. Centrifuge at 5000–10,000 × g (usually 8000 × g), 30–60 min, 22–25°C (room temperature).
4. After centrifugation, slowly transfer the supernatant into a clean 2/3/5 L media bottle.
5. Separate the centrifuge bottles and caps and decontaminate them by Route A or Route B (see "Decontamination of Reusable Bottles, Tubes and Glassware"), then wash.

Expected result: Clear supernatant free of bacterial pellet, ready for filtration.

⏱ This step will take approximately 1.5 h (including loading, balancing, centrifugation, and supernatant transfer). If you cannot finish on the same day, the supernatant can be kept at 4°C and the protocol continued the next day.

Step 2: Filtration (Removal of Residual Bacteria and Debris)

Note: Traditional dead-end filtration (sequentially using 0.8, 0.45, and 0.2 µm filters) can be used; for larger volumes, a tangential flow filtration (TFF) system is recommended to accelerate processing. However, not all labs have the 0.2 µm cassette.

Sequential Dead-End Filtration

1. Assemble the vacuum filtration unit: fit a 47 mm 0.8 µm PES membrane into the filter holder and connect the filtration flask to the vacuum pump. The membranes do not need to be pre-wetted.
2. Pour the supernatant from Step 1 into the funnel and filter it under vacuum. Replace the membrane whenever the flow rate drops noticeably.
3. Repeat the filtration with a fresh 47 mm 0.45 µm PES membrane, then with a fresh 47 mm 0.2 µm PES membrane, collecting the filtrate each time. Use a new membrane at every stage.
4. Discard the used membranes as biological waste and send the filtration glassware for sterilization and washing.

Tangential Flow Filtration (TFF) — Sartorius 0.2 µm cassette

1. Rinse the internal system of the TFF equipment with 2 L of DI water to remove residual liquid.
2. Insert the feed tube into the sample bottle. Collect the permeate; route the retentate back into the feed tube to circulate until all liquid is collected as permeate.
3. Flush the lines with 2 L of DI water, then circulate 1 L of 0.01 M NaOH to inhibit bacterial growth for storage.
4. Decontaminate the original sample bottles by Route A or Route B (see "Decontamination of Reusable Bottles, Tubes and Glassware"), then wash.

Expected result: Filtrate free of residual bacteria and debris (clarified through 0.2 µm).

⏱ This step will take approximately 1–2 h, depending on sample volume and filter flux. If you cannot finish on the same day, the filtrate can be kept at 4°C and the protocol continued the next day.

Purpose of Steps 1–2: to remove bacteria and large particles/debris from the sample.

Step 3: Concentration (Volume Reduction)

Two methods can be used for this step. Both give an equivalent result — use whichever system is available:

Method A — Tube-based centrifugal concentration (Amicon® Ultra-15, 100 kDa MWCO)

Performed in E29 (building 4.1, UFZ) using an Eppendorf centrifuge and 100 kDa Millipore (Amicon® Ultra-15) concentrator tubes.

1. Rinse the 15 mL concentrator tubes with DI water and place the filter insert into the plastic tube.
2. Use a pipette to transfer the filtered sample into the concentrator tubes (max capacity per run: 16 tubes × 15 mL = 240 mL).
3. Centrifuge at 4°C, 2000 × g, 3–5 min (adjust the time based on the residual volume in individual tubes).
4. When the retentate volume is <1 mL, transfer it to a 50 mL centrifuge tube for pooling. Repeat this process until all samples are concentrated (est. time: ~2 hours for 2 L).
5. After the last collection, rinse the concentrator tubes with 1–20 mL of PBS to recover vesicles attached to the membrane and add the rinse to the pooled sample. The pooled volume does not have to fill the ultracentrifuge tubes — they are topped up with PBS in Step 4.2.
6. Store the concentrated samples at 4°C.
7. Rinse the filter inserts multiple times with DI water and store them in 0.01 M NaOH or 75% ethanol; wash and dry the plastic outer tubes.

Expected result: Retentate volume suitable for loading into the ultracentrifuge tubes in the next step.

⏱ Method A will take approximately 2 h for 2 L of sample. If you cannot finish on the same day, the concentrated sample can be kept at 4°C and the protocol continued the next day.

Method B — Tangential Flow Filtration (TFF) concentration (Millipore 100 kDa TFF system)

Use this method when a dedicated 100 kDa TFF concentrator system is available; it is faster than Method A for large volumes.

1. Rinse the TFF concentration system with DI water.
2. Load the filtered sample (from Step 2) into the TFF system reservoir.
3. Circulate the sample through the 100 kDa cassette, discarding the permeate, until ~150 mL of concentrated retentate remains.
4. Rinse the remaining sample out of the system with PBS and collect a further ~80 mL, giving ~230 mL of pooled retentate in total.
5. Flush the system with DI water, then circulate 0.01 M NaOH for storage between uses (as in Step 2).
6. Store the pooled retentate at 4°C until ultracentrifugation.

Expected result: Same as Method A, achieved more quickly.

⏱ Method B will take approximately 1 h for 2 L of sample (faster than Method A). If you cannot finish on the same day, the concentrated sample can be kept at 4°C and the protocol continued the next day.

Typical final volume prior to ultracentrifugation (either method): up to 6 × 38.5 mL (38.5 mL = capacity of each ultracentrifuge tube; up to 6 tubes/samples can be processed per ultracentrifugation run).

Step 4: Ultracentrifugation (EV Enrichment)

1. Filter the concentrated sample using a 0.2 µm syringe filter and transfer it into the 38.5 mL ultracentrifuge tubes.
2. Top up each tube with pre-cooled PBS to ~37–38 mL, close to the 38.5 mL capacity. Underfilled tubes collapse under vacuum — see Critical Control Points.
3. Balance the ultracentrifuge tubes precisely to 0.01 g using an analytical balance.
4. Place the tubes symmetrically into the SW 32 Ti rotor buckets (e.g., #1 opposite #4, #2 opposite #5). All six buckets must be attached to the rotor for every run, even when fewer tubes are loaded.
5. Set parameters: Rotor/Tube: SW 32 Ti / 38.5 mL; Speed: 100,000 × g; Time: 2 h; Temperature: 4°C; Accel: Max; Decel: 7/8 (or set to no coast).

6. Operation monitoring and run: click “Start vacuum” to evacuate the chamber; the run begins after 2–3 min and the rotor reaches the set speed in ~5 min. After the run, the vacuum is released once the speed drops below 2000 rpm; then click “Open vacuum” before removing the tubes.
7. Remove the tubes and pour off the supernatant completely in one smooth motion. Add 100–200 µL of pre-cooled PBS to the tube and resuspend the pellet by pipetting up and down repeatedly aimed at the bottom (final total volume should be ~1 mL). If more than one tube was run, pool the resuspended pellets from all tubes into a single collection tube and adjust the combined volume to ~1 mL with pre-cooled PBS.
8. Clean the ultracentrifuge tubes for reuse: decontaminate them by Route A or Route B (see “Decontamination of Reusable Bottles, Tubes and Glassware”), then rinse with DI water, dry and store at room temperature. Thin-wall polypropylene tubes tolerate only a limited number of autoclave cycles, so Route B is preferred for tubes that are reused often.

Expected result: EV pellet resuspended in ~1 mL PBS, ready for downstream storage or assays.

⏱ This step will take approximately 3–3.5 h (including tube loading, balancing, the 2 h run, and pellet resuspension). If you cannot start the ultracentrifugation run on the same day, the pre-filtered concentrated sample can be kept at 4 °C and the run continued the next day.

Purpose of Steps 3–4: to enrich the vesicles.

In theory, EVs are obtained once the steps above are complete. Depending on your experimental design, however, consider whether further purification is needed (see Optional Step below).

Optional Step: Density Gradient Purification (Further Purification)

If downstream analysis involves proteomics, separate and purify using a sucrose solution or iodixanol. This step is generally not required for functional validation assays. This procedure is documented separately as Protocol EV-Purification-001 (Density Gradient Purification).

Critical Control Points

⚠ Step	Why it is critical	How to avoid mistakes
Step 4.1–4.2 — Loading and filling the ultracentrifuge tube	Tubes are thin-wall plastic with low resistance to compression; if the liquid volume is not nearly full, the vacuum drawn during the run will collapse/destroy the tube (see Figure 2)	Fill the tube close to its capacity (e.g., top up to 37–38 mL with PBS for a ~30 mL sample); verify the volume before loading
Step 1.2 / Step 4.3 — Balancing bottles/tubes	Imbalance can damage the rotor and compromise run safety	Use a scale or match liquid meniscus for centrifuge bottles; balance ultracentrifuge tubes precisely to 0.01 g with an analytical balance; load rotor positions symmetrically

Troubleshooting

Problem	Possible Cause	Solution
Ultracentrifuge tube collapsed/deformed after the run (Step 4)	Combined sample + PBS volume in the tube was insufficient; the resulting air gap allows the vacuum drawn during the run to crush the thin-wall tube	Before the run, top up the liquid volume close to the tube's capacity (e.g., to ~37–38 mL for a 38.5 mL tube); double-check the volume before loading; ensure precise balancing
No visible pellet after ultracentrifugation	Normal in most runs — bacterial EV pellets are small and translucent and are often not visible by eye	Do not discard the tube. Resuspend the bottom of the tube as described in Step 4.7 and confirm recovery by NTA before concluding that the run failed

Problem	Possible Cause	Solution
TFF flux drops sharply during filtration or concentration	Membrane fouling by residual cells and debris	Pause and rinse or backflush the cassette with DI water; confirm that the Step 1 centrifugation removed the cell pellet completely; replace the cassette if the flux does not recover
Filtrate still turbid after 0.2 µm filtration	Membrane damaged or bypassed through a poor seal; incomplete bacterial removal in Step 1	Re-filter through a fresh 0.2 µm membrane; check the seal of the filter holder; check the Step 1 supernatant for carry-over of the pellet

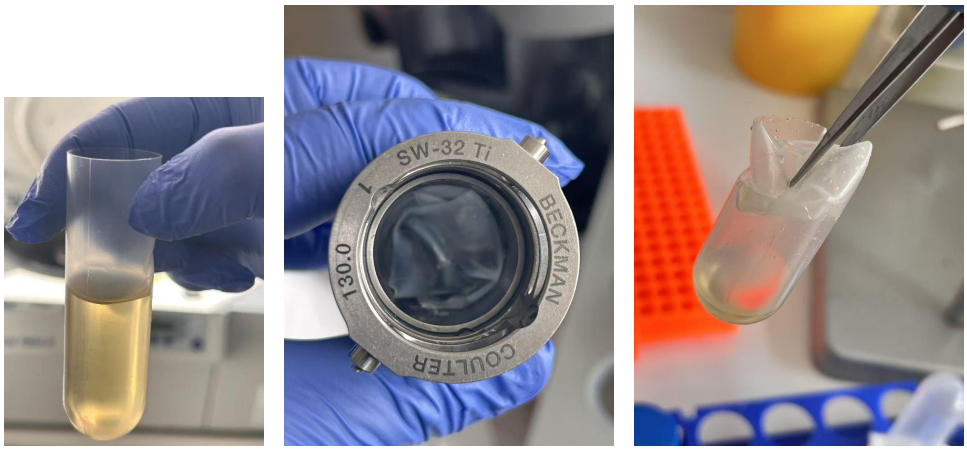


Figure 2. Vacuum-induced collapse/deformation of the ultracentrifuge tube

Left to right: (1) insufficient liquid volume in the tube ("the volume is not enough"); (2) a visibly deformed tube inside the rotor bucket; (3) close-up of the crushed tube.

Expected Outcome

A successful preparation yields a translucent to pale-yellow EV pellet resuspended in ~1 mL of pre-cooled PBS, with no visible bacterial contamination, suitable for downstream functional validation assays. Further purification by density gradient can be performed if proteomic analysis is needed.

Storage and Stability

Item	Storage Condition	Stability
Concentrated sample (pre-ultracentrifugation)	4°C	1–2 days
Final EV pellet/resuspension	4°C (short term) or −20°C/−80°C (long term)	4°C — short-term storage (~1 week). −20°C or −80°C — long-term storage; aliquot before freezing and avoid repeated freeze–thaw cycles. Choose the temperature based on the intended downstream experiment
TFF system (idle/storage)	Internally circulated with 1 L of 0.01 M NaOH	Inhibits bacterial growth
Concentrator filter inserts	Stored in 0.01 M NaOH or 75% ethanol	—

References

SOPs

- EV-Purification-001 — Density Gradient Purification (Optional Step)

Publications

Su, Z. Z., Hou, Y., Lin, J. J., et al. (2026). *Extracellular vesicles from biofilm and planktonic Pseudomonas aeruginosa: proteomic profiles, iron chelation and functional implications*. *Archives of Microbiology*, 208, 239. https://doi.org/10.1007/s00203-026-04794-w — detailed methods are described in this publication.

Zhu, L.-T., Huang, H.-N., Avellán-Llaguno, R. D., Qin, Y., An, X.-L., Su, J.-Q., Huang, Q., & Zhu, Y.-G. (2022). *Diverse functional genes harboured in extracellular vesicles from environmental and human microbiota*. *Journal of Extracellular Vesicles*, 11, e12292. https://doi.org/10.1002/jev2.12292

Instrument manuals

- *Beckman Coulter SW 32 Ti rotor manual*
- *TFF system manual*
- *Amicon® Ultra-15 concentrator tube instructions*

Ordering Information

Consumables used in this protocol and where they are ordered.

Table 5. Consumables and ordering information

Item	Specification	Supplier / Cat. No.	Price
Ultracentrifuge tubes	Open-Top Thinwall PP, 38.5 mL, 25 × 89 mm	Beckman Coulter, Cat. No. 326823 (50/pack, non-sterile) Sterile version: Beckman C14301 = Fisher NC2259182 (48/pack)	326823: on request C14301: US$ 567.12 / 48
Concentrator tubes	Amicon® Ultra-15 centrifugal filter, 100 kDa MWCO	Merck Millipore, UFC9100 — pack sizes 8 / 24 / 96 24-pack = UFC910024	On request (sign-in)
Syringe filters	Millex®-GP, 0.22 µm PES, sterile, 33 mm	Merck/Sigma-Aldrich, Cat. No. SLGPR33RS	€ 239.00 / 50
Filter membrane — 0.8 µm (pre-filter)	PES, 47 mm	Cytiva (Pall) Supor®-800, Cat. No. 60110 — Fisher 17184481 (100/pack)	On request
Filter membrane — 0.45 µm	PES, 47 mm	Cytiva (Pall) Supor®-450, Cat. No. 60173 — Fisher 17134491 (100/pack) Alternative: Sartorius type 15406, 15406--47-----N	≈ SEK 3,125 / 100 (Fisher SE list; DE price on request)
Filter membrane — 0.2 µm	PES, 47 mm	Cytiva (Pall) Supor®-200, Cat. No. 60301 — Fisher 17164501 (100/pack) Alternative: Sartorius type 15407, 15407--47-----MIN	On request
Syringes	20 mL / 50 mL, sterile	Any supplier	—
1 L centrifuge bottles	for SORVALL LYNX 6000	Thermo Scientific	—

Item	Specification	Supplier / Cat. No.	Price
Centrifuge tubes	50 mL	Any supplier	—
EP tubes	1.5 mL / 2 mL	Eppendorf Safe-Lock Tubes	—
TFF cassette (Step 2 — 0.2 µm filtration)	Hydrosart®, 0.2 µm, 0.1 m²	Sartorius Sartocon® Slice, Order No. 3051860701W	On request Around 1,500 Euro
TFF cassette (Step 3, Method B — 100 kDa)	Pellicon® 2 Mini ultrafiltration module, 0.1 m², 100 kDa NMWCO, with Pellicon® Mini cassette holder	Merck: holder XX42PMINI, module P2C100C01 Sartorius alternative: Sartocon® Hydrosart® 100 kDa, 0.1 m², E-Screen, 3051446801E	Merck: on request Sartorius: US$ 1,610
Pump (Step 3, Method B)	Pump head YZ15; takes #16 and #17 tubing	Golander BT100S variable-speed peristaltic pump, 1–4 channels, 0.0001–720 mL/min	—
Wofasteril	peracetic acid; used as a 2% (v/v) working solution	Any supplier	—
PBS salts	NaCl, KCl, Na_2HPO_4 (anhydrous), KH_2PO_4 — analytical grade	Any supplier	—
NaOH	pellets, analytical grade	Any supplier	—
Ethanol	70–75% (v/v), lab-grade	Any supplier	—

List prices checked on 2026-10-22, excluding VAT. Prices depend on the framework contract and on the pack size ordered — confirm them before ordering. Cells highlighted in yellow still need to be checked.

Revision History

Version	Date	Changes	Author
1.0	2026-10-21	Initial version	Haining Huang